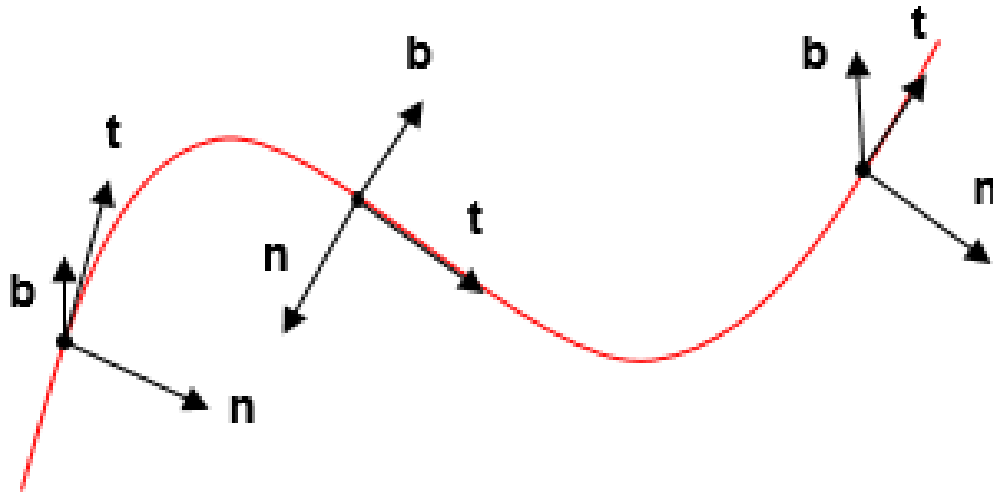


TNB – Frames



$T(t)$: (**Unit Tangent**, Direction, nose of the plane)

$$T(t) = \frac{\vec{r}'(t)}{\|\vec{r}'(t)\|}$$

$N(t)$: (**Unit Normal**, Turn, tailfin of the plane)

$$N(t) = \frac{T'(t)}{\|T'(t)\|}$$

$B(t)$: (**Unit Binormal**, twist, wings of the plane)

$$B(t) = T(t) \times N(t)$$